IMPULSE: Quantifying the Real-Data Limits of Passive Orbital Sensing for Sub-Catalog Debris-Environment Model Discrimination

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Abstract

The sub-catalog (≤ 10 cm) orbital debris environment is poorly constrained; NASA's ORDEM 3.1 and ESA's MASTER 8 diverge by over two orders of magnitude in critical LEO regimes. IMPULSE tests whether orbits of decommissioned, non-maneuvering satellites contain tiny signals of microdebris impacts that could discriminate between the models. Historical TLE data are processed with a 9-state augmented Extended Kalman Filter and Fraser-Potter smoother (non-singular (h, k) eccentricity) to flag statistically significant orbital deviations as candidate impulses. Measurement covariance is estimated per object from its own TLE stream. Validation sets the design threshold $\eta = 417.94$ (99.9th percentile of 18,582 synthetic-null trials) and certifies 50% detection at $\Delta v_{50} = 1.19$ m/s (1.21 m/s on real TLEs). Applied to 3,325 spacecraft ($15,760$ windows), the pipeline yields 6 candidates against 15.76 expected from the design noise rate: the count lies below that expectation ($P \approx 0.005$) yet above what the more realistic permuted null predicts, so no positive excess is certifiable against either background and the one-sided 95% Poisson upper limit, conditional on the design background, is zero. Candidates carry 1.5–15 m/s impulses, compatible only with 3–6 cm head-on strikes. In the decision-relevant 1–10 cm band the models agree within a factor of ~ 3 and predict essentially no events; zero detections is therefore fully consistent. IMPULSE demonstrates fleet-scale synthetic sensitivity and places a weak bound (≈ 280–$570\times$ the model rate) on the $\geq$ 2–3 cm impact rate that reflects present performance limits rather than discriminating the models.

1 Introduction

1.1 Sub-Catalog Orbital Debris

Sub-catalog debris is the population of space debris too small to be tracked and logged in official space object directories, with the line typically drawn around 10 cm (Liou, 2020; Manis et al., 2023), the threshold that defines IMPULSE's regime of relevance. Radar and optical space-surveillance approaches cannot maintain a continuous catalog at this scale due to their sensitivity and noise floors (Manis et al., 2023). Yet there is far more sub-catalog debris than cataloged debris, and cumulative tiny hypervelocity impacts could change orbital trajectories and degrade spacecraft surfaces (Murray et al., 2023; Sagnières & Sharf, 2017); in rarer cases, larger debris could damage critical subsystems.

1.2 Existing Debris-Environment Models

Since this population cannot be directly observed, statistical characterization is necessary. The two major debris-environment models are NASA's Orbital Debris Engineering Model (ORDEM 3.1) and ESA's Meteoroid and Space Debris Terrestrial Environment Reference (MASTER 8) (Braun, Horstmann, et al., 2021; Matney et al., 2019). Built from different sources and assumptions, they agree in the cataloged population but diverge dramatically in the sub-catalog regime, up to $\sim$138$\times$ in the regime IMPULSE operates in (Horstmann et al., 2021). Because these models quantify a debris population dangerous enough to be considered in spacecraft mission design, weighing one over the other could mean exaggerated mission costs or the underprotection of vital subcomponents (Christiansen, 2009). Independent validations are therefore necessary, but existing attempts are limited: returned surfaces provide ground truth only at specific epochs (Stansbery & Foster, 2005); dedicated detectors like DEBIE have excellent time resolution but are limited by trajectory, cost, and rarity (Kuitunen et al., 2001); and optical or radar approaches are weather-dependent, geographically restricted, and noise-floor limited (Manis et al., 2023).

1.3 Passive Orbital Sensing and Research Questions

A different, comparatively less explored option is to study the past orbital behaviour of spacecraft that already exist: instead of directly observing debris, this approach looks for debris-induced perturbations in orbital histories. Existing work has approached orbital anomalies as individual discrete events, testing detection of synthetically injected impulses within osculating elements (Bennett et al., 2023). TLE histories have likewise been mined for maneuvers (Lemmens & Krag, 2014) and their population-wide orbit errors characterized (Flohrer et al., 2008); IMPULSE extends the same historical asset to the much weaker signature of sub-catalog debris impacts. IMPULSE builds on that foundation but shifts the problem into the regime that actually exists in the historical record: real TLE histories expressed in mean elements (Hoots & Roehrich, 1980). This introduces a substantially different measurement-error environment, catalog artifacts, and force-model mismatches absent from purely synthetic experiments, with the measurement error of each stream characterized from the stream itself (Section 3.2). The present work is a first attempt to test whether the passive-sensing concept remains viable once these real-world complications are included.

To isolate candidate perturbations from maneuvers, IMPULSE uses TLE histories of decommissioned satellites, whose observed orbital evolution can be modeled primarily through natural perturbations and

measurement effects. A fleet of these satellites provides far more statistical information than one spacecraft. IMPULSE is therefore not a tool to reconstruct the debris population, but a discrimination study between ORDEM and MASTER. The primary question is: to what extent can historical TLE data from decommissioned, non-maneuvering LEO satellites serve as passive measurements of sub-catalog debris impacts, and do the measurements have enough statistical power to discriminate between the models' predictions? The hypothesis is that debris impacts impart enough impulse to produce small transient orbital-state perturbations detectable in TLE-derived parameters (Bennett et al., 2023); whether such perturbations can be distinguished from catalog, measurement, and propagation-model discrepancies, including the TLE mean-element convention mismatch, is the central experimental question. The main practical difficulty is unmodeled perturbations not accounted for in the model or the background subtraction: outside of drag, solar radiation pressure, and geomagnetic storms, this primarily includes TLE-cataloging error such as discontinuities and station changing (Osweiler, 2006; Vallado & Cefola, 2012). The present work therefore (i) calibrates the measurement covariance per object from each stream's own residuals, (ii) quantifies the convention mismatch with an end-to-end synthetic experiment, and (iii) derives the fleet-scale trigger rate against a null constructed from the real streams themselves.

2 Experimental Design

2.1 Fleet Construction and Data

The concept relies on non-maneuverability: decommissioned spacecraft do not fire correction or attitude control burns. At LEO (700–1000 km), drag is substantially weaker than at lower altitudes, allowing orbital histories to persist over long timescales (Anz-Meador & Liou, 2010; Liou & Johnson, 2008). The 700–1000 km band was selected to ensure exposure to the dense microdebris population from the Fengyun-1C breakup and the Kosmos-Iridium collision belt (Anz-Meador & Liou, 2010; Liou & Johnson, 2008; Pardini & Anselmo, 2009); its low drag lets debris persist longer, extending the usable history, and its heavy use of Sun Synchronous Orbit (SSO) makes it a region of interest (Schild & Noomen, 2022).

The initial fleet consisted of all decommissioned spacecraft with Space-Track TLE coverage between 2009 and 2026 whose mean altitude lay inside the band; the 2009 cut-off denotes the post-Fengyun-1C breakup event, which dramatically changes the flux in the region. Operational or station-keeping objects were removed, with decommissioned status determined from Space-Track operational-status flags (Li & Chen, 2018) cross-checked against the absence of station-keeping behavior. Rocket bodies and cataloged debris fragments were excluded by classification. The final catalog contained 3,325 decommissioned spacecraft.

Each object's TLE history was segmented into successive, non-overlapping 90-day windows. Four quality cuts ensured numerical and statistical stability:

- Minimum of 15 TLEs inside the window.
- Data discontinuities trimmed out.
- Altitude changes inside the window must be less than 10 km between TLEs, indicating station-changing or other cataloging artifacts.

- Exclusion of the first window for objects whose total span exceeds 90 days, to account for filter transients.

Of the original 22,888 raw windows, 15,830 (69.2%) survived the cuts. A 7-day edge trim suppresses filter transients and smoother boundary effects, leaving around 76 usable days per window. A further symmetric exclusion removes windows in which the backward pass diverges (non-finite state or covariance, or a diagonal covariance exceeding 10^{12}), identically for the fleet and every null population (Section 4.3): 70 fleet windows (0.44%) are excluded, leaving 15,760 valid fleet windows, and all rate denominators below use post-exclusion counts. Each accepted window contributes a binary trigger outcome; these are aggregated across the fleet into a population-level trigger rate, converting many individually weak measurements into a single statistical measurement of the underlying debris environment.

2.2 IMPULSE Architecture and Decision Framework

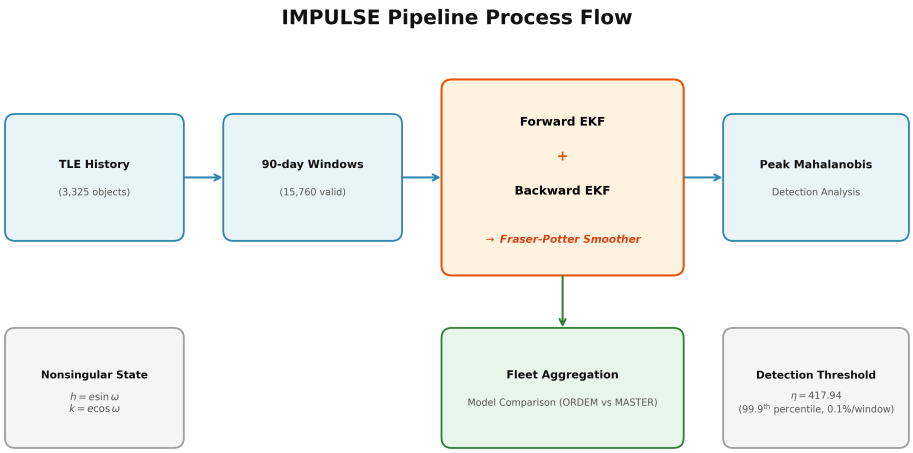


Figure 1: The IMPULSE pipeline. TLE histories for 3,325 objects are segmented into 90-day windows. A per-object measurement covariance is estimated from each stream's own one-step prediction residuals before filtering. Each window is processed by a forward EKF, backward EKF, and Fraser-Potter smoother. The peak Mahalanobis distance is tested against the calibrated threshold $\eta = 417.94$, the 99.9th percentile of the pooled synthetic null (18,582 valid trials; 0.1% false-alarm rate per window). Detections are aggregated across the fleet for model comparison. [AI-generated graphic using Claude (Anthropic), subsequently modified by the student researcher, 2026]

As summarized in Figure 1, each window is run through a 9-state augmented Extended Kalman Filter (EKF) with a non-singular eccentricity-perigee element pair, using a per-object measurement covariance estimated from the object's own stream, and smoothed via a fixed-interval Fraser-Potter smoother with first-order Gauss Markov dynamic model compensation. Candidate impulses are decided by a calibrated threshold on the residuals, and the candidate events are aggregated across the fleet. A validation framework provides the calibrations, thresholds, and false-positive characterization that adjust this raw count to a background-subtracted count, which is compared to the expected ORDEM/MASTER event counts. Detection count alone does not constitute a successful measurement: a large portion of the triggers could indicate an inadequately characterized background, an overly sensitive threshold, or systematic error. The experiment

succeeds only if the background-subtracted signal is statistically distinguishable from the quantified false-positive distribution and contains sufficient information to discriminate between ORDEM and MASTER predictions.

3 Orbital-State Estimation

3.1 Orbital-State Representation and Dynamical Model

IMPULSE represents the orbital state in a nine-element set with a nonsingular eccentricity-perigee pair, rather than classical Keplerian elements (Broucke & Cefola, 1972). The six orbital elements are

$$\mathbf{x}_{\text{orb}} = (a, h, k, i, \Omega, M) \tag{1}$$

where $h = e\sin\omega$ and $k = e\cos\omega$. The transformation replaces only the eccentricity-argument-of-perigee pair with variables that remain well behaved as eccentricity approaches zero. The state is augmented with three RSW acceleration components via a first-order Gauss-Markov (FOGM) process that absorbs residual, unmodeled accelerations, tuned primarily to mitigate long-term accumulating effects such as atmospheric-density and ballistic coefficient mismatch:

$$\mathbf{X} = [a, h, k, i, \Omega, M, W_R, W_S, W_W]^T \tag{2}$$

The (h, k) substitution is required by the orbital regime itself: in the classical Gauss variational equations, $d\omega/dt$ contains a factor $1/e$ (Vallado, 2022). As drag circularizes orbits in the SSO regime, e approaches 0, routinely falling below 10^{-4}–10^{-5}, at which point the classical formulation becomes numerically unstable while the nonsingular (h, k) rates remain finite and well-conditioned. Two near-singular terms nevertheless remain: the mean-anomaly rate retains a $1/e$ coefficient, floored at 10^{-8} and evaluated at the true eccentricity for all fleet orbits ($e \gtrsim 10^{-5}$), and the nodal term $1/\sin i$, inactive for a sun-synchronous-dominated fleet. Numerical stability therefore rests on the complete propagated equations and detector: finite peaks were computed for 15,760 of 15,830 fleet windows, and the 18,582-trial null distribution is smooth and bounded.

The dynamical model includes the dominant environmental accelerations evaluable through publicly available indices and ephemeris: Earth's gravity through J_2 (Brouwer, 1959); atmospheric drag with stochastic C_d driven by an Ornstein-Uhlenbeck process, piecewise-fit against NRLMSISE00 (Picone et al., 2002; Schiemenz et al., 2020; Uhlenbeck & Ornstein, 1930); solar-radiation pressure via a cannon-ball model (McMahon & Scheeres, 2015); and lunisolar third-body perturbations (Kaula, 1962). The stochastic C_d is a parameter of the drag model, not an element of the nine-state vector; all other unmodeled accelerations (higher-degree gravity, albedo, thermal radiation) are absorbed into the FOGM process-noise terms. Measurement noise covariances are anchored to published TLE accuracy statistics (Osweiler, 2006), which serve as a per-channel floor on the per-object empirical covariance of Section 3.2, and are treated as diagonal, a noteworthy limitation partly mitigated by the FOGM and the smoother. This combination of nonsingular elements, a physically motivated but incomplete force model, and a low-order stochastic acceleration process is the null model against which candidate impulses are later tested.

3.1.1 Note on TLE Element Convention

The observations originate from published TLE fields rather than independently determined orbital states; the initializing semi-major axis is recovered from the TLE mean motion as $a = (\mu/n^2)^{1/3}$, and no other convention correction is applied. The estimator propagates an orbit-averaged secular state (J_2 secular rates, drag, SRP, and lunisolar rates). While both the observations and the estimator state are expressed in mean-element form, their conventions are not assumed identical: TLE parameters are typically associated with SGP4 theory (Hoots & Roehrich, 1980), so convention differences may contribute temporally correlated residuals. This contribution is quantified directly in Section 5.1.

3.2 Measurement Model and Covariance

The measurement at epoch t_k is the six-element vector derived from the TLE fields,

$$\mathbf{z}_k = [a, e, i, \Omega, \omega, M]_k^\top + \boldsymbol{\epsilon}_k, \tag{3}$$

with $\boldsymbol{\epsilon}_k$ modeled as zero-mean Gaussian noise with diagonal covariance R in classical element coordinates. Before the update, (e, ω) is transformed into (h, k), giving $\tilde{z}_k = HX_k + \tilde{\varepsilon}_k$ with $H = [I_{6\times6} \mid 0_{6\times3}]$. As the transformation is nonlinear, the covariance is transformed using the local Jacobian: the variances used for h and k are $R_h = \sin^2\omega R_e + e^2\cos^2\omega R_\omega$ and $R_k = \cos^2\omega R_e + e^2\sin^2\omega R_\omega$.

Published TLE accuracy statistics are population-wide: they describe the catalog average, not any individual stream. IMPULSE therefore estimates the operating measurement covariance R per object, from the TLE stream itself. For each consecutive pair of TLEs, the previous element set is advanced to the next TLE epoch with the Brouwer secular rates; the wrapped residual of the next TLE against that prediction is exactly the one-step innovation the filter faces. The per-channel scatter of those residuals is the covariance the stream actually exhibits, as opposed to what the catalog on average suggests it should exhibit. Because a real maneuver is itself a genuine one-step residual, the scatter is estimated from the median of squared residuals, so a real impulse widens R only mildly instead of blinding the detector. The per-object covariance is floored per channel at the published statistics (the detector is never more confident than the published accuracy) and capped at $100\times$ for pathological objects; the resulting R is applied to the forward update, the backward update, and the initial covariance. Across the fleet, the median object sits near the published floor on most channels (a factor of 1.0 on a, e, i, Ω, and ω, and 1.5 on M); the widening acts on a minority of noisier objects, which is where the false positives originate (Figure 6). The diagonal R is an approximation: the per-channel form cannot represent cross-channel correlation, notably between a and M, which bounds what the per-channel calibration can absorb.

3.3 Forward and Backward Extended Kalman Filtering

Isolating candidate impulsive events within a discrete, noisy mean-element trajectory requires sequential estimation, performed here with two Extended Kalman Filters that operate on the same 90-day window in opposite directions (Jazwinski, 1970; Kalman, 1960). The forward filter is initialized at the first valid TLE and marched forward: at each measurement epoch it predicts the nine-element state under the null dynamical model, computes the innovation, and updates the state estimate and covariance; process-noise intensity is set by the FOGM parameters τ and q so that slowly varying modeling errors are absorbed rather than falsely

detected. Once the forward pass ends, a second filter is initialized at the terminal state and covariance and run backward, incorporating future states unavailable to the forward pass. The forward filter can detect an abrupt change only after it has occurred and only relative to the preceding data; the backward filter sees the same change relative to the subsequent data. Their mutual inconsistency, normalized by joint uncertainty, is therefore a sensitive indicator of a local dynamical discontinuity, and becomes the detection statistic.

3.4 Fraser-Potter Smoothing and Process Noise

The detection statistic must isolate a localized dynamical inconsistency, which requires both past and future information at each epoch. IMPULSE therefore fuses the two EKF passes with a Fraser-Potter smoother (Fraser & Potter, 1969). At each interior epoch k, the forward a-posteriori estimate is conditioned on measurements $z_1 \dots z_k$ and the backward a-priori estimate on $z_{k+1} \dots z_N$; each measurement enters the fusion exactly once, as required by the Fraser-Potter decomposition. The smoothed state is

$$\mathbf{x}_S = W\mathbf{x}_F + (I - W)\mathbf{x}_B, \qquad W = P_B(P_F + P_B)^{-1} \tag{4}$$

where subscripts F and B denote the forward a posteriori and backward a priori quantities. The three unmodeled-acceleration states are driven by the FOGM process

$$\dot{\mathbf{w}} = -\frac{\mathbf{w}}{\tau} + \boldsymbol{\nu}, \qquad \langle \boldsymbol{\nu}(t)\boldsymbol{\nu}(t')^\top \rangle = q, \delta(t - t'), \mathbf{I} \tag{5}$$

with fixed parameters $\tau = 3 \times 10^4$ s and $q = 10^{-19}$, set by the validation ladder so that slow model mismatch is absorbed rather than interpreted as impulsive events (Jazwinski, 1970; Schiemenz et al., 2020; Uhlenbeck & Ornstein, 1930). The resulting detection statistic is the peak Mahalanobis distance between the forward and backward estimates, evaluated with the smoother covariance (Bennett et al., 2023).

Calibration limitations of the estimator. The detector uses a scalar anomaly score rather than an analytically standardized test statistic: the smoothed covariance P_S is that of the smoothed state, not of the discrepancy $\Delta = x_F - x_B$ itself, so no analytic distribution is assumed. The backward filter is initialized at the terminal forward estimate with covariance inflated by a factor of 100, which reduces but does not fully eliminate shared information between the two states; the initialized cross-covariance is not modeled, and the reverse-time propagation of the FOGM states is likewise an approximation. The threshold η is therefore defined empirically as a percentile of the null distribution produced by this identical estimator and peak statistic, so any correlation-induced or statistic-level bias is common to calibration and evaluation and is absorbed by the empirical calibration rather than bounded analytically. The "Mahalanobis" label is likewise a convenience, not a claim of standardization: the score is a univariate, smoother-covariance-normalized semi-major-axis difference, so magnitudes of several hundred are indices relative to the empirical null rather than physically meaningful Gaussian sigmas; the principled generalization — the full quadratic form of the six-state forward–backward discrepancy under its own covariance — requires cross-covariance terms the smoother does not carry, and the scalar score is retained because the entire calibration and null construction are defined on it. Because the fleet statistic takes its maximum over the trimmed window interior, a subset of the full-window search domain used in calibration, the calibrated percentile is conservative for the evaluated

statistic. These limits are discussed once, in Section 7.2.

3.5 Candidate Impulse Detection

The detection statistic is the peak Mahalanobis distance between the forward and backward state estimates, evaluated with the smoother covariance:

$$s_k = \frac{\left|a_{F,k} - a_{B,k}\right|}{\sqrt{P_{S,aa,k}}}, \qquad S_{\text{window}} = \max_{k \in \mathcal{W}} s_k, \tag{6}$$

where $a_{F,k}$ and $a_{B,k}$ are the forward a-posteriori and backward a-priori semi-major axes at epoch k, $P_{S,aa,k}$ is the corresponding smoothed covariance element, and $\mathcal{W}$ denotes the epochs remaining after the 7-day edge trim. The statistic is evaluated on the semi-major-axis component alone: it is the channel to which injected impulses impart their largest persistent signature, and η, the injection-recovery curves, the null tests, and all fleet statistics are computed from this same scalar definition. Restricting to this channel does not by itself establish that covariance approximations in the remaining components cannot influence it: the smoother couples channels through the gain, and any such influence is absorbed into the empirical calibration. Three intertwined claims must be clarified: a **statistical trigger** is a window whose $S_{\text{window}} > \eta$; a **recovered synthetic impulse** is a trigger that coincides with a known injected Δv (calibration tests only); and a **confirmed debris impact** would require additional post-processing or attribution isolating the event from non-debris systematic noise. The present work reports only the first; the background-subtraction and model-comparison stages (Sections 5–6) bound how much of the trigger population can be attributed to debris.

4 Validation and Calibration

4.1 Validation Strategy

A statistical trigger is a candidate orbital perturbation, not automatically debris: estimator transients, force-model incompleteness, unmodeled physical effects, and TLE-specific artifacts can all generate statistically significant residuals. IMPULSE therefore runs the staged validation ladder of Table 1, progressing from fully controlled synthetic conditions to real TLE data.

Gate	Question	Purpose
1. Convention-consistency null	Does the filter trigger on known truth in the TLE convention?	Bound intrinsic false-positive rate
2. Bias recovery	Can it track slowly varying forces?	Verify FOGM absorbs model error
3. Confounders	Can density, C_d, or K_p mimic impacts?	Quantify confounder FP rate
4. Strike injection	Can it recover known impulses?	Measure detection efficiency
4b. Real-TLE injection	Does sensitivity persist on real TLEs?	Test under observational noise

Table 1: Validation ladder. [Table created by the student author using LaTeX, 2026]

4.2 Synthetic Nulls and Physical Confounders

The intrinsic null measures the rate at which the estimator triggers on synthetic truth carried through the full TLE-like observation pipeline (Section 5.1): 0.0% (0 of 7000 windows) at the adopted threshold η, with mean peak statistic ≈0.8. The confounder null deliberately injects density variations, C_d drift, and geomagnetic activity, testing $F_{10.7}$, K_p, and nominal C_d separately and together by holding two at a degraded baseline ($F_{10.7} = 150.0$, $K_p = 2.0$, $C_d = 2.2$) and letting the other influence the pipeline. When realistic but non-impact perturbations were injected into a synthetically clean null, the false-positive rate remained 0.0% (zero triggers across all five configurations, 32,646 windows; Table 2). These low rates confirm that the pipeline does not independently invent detections under controlled conditions: the force model is effectively absorbed by the FOGM, and no major model noise floor is evident in the synthetic tests; residual errors likely originate in real-TLE data quality and other systematics instead.

Filter assumption	**Windows**	**Triggers**	**FP rate (%)**	**Mean peak D_M**
Matched baseline	6,526	0	0	5.522
Flat $F_{10.7}$	6,528	0	0	5.534
Quiet K_p	6,532	0	0	5.467
Flat environment (all)	6,529	0	0	5.508
Nominal C_d	6,531	0	0	5.531

Table 2: Confounder test results summary. All five configurations produced zero triggers; mean peak statistics (5.47–5.53) sit below the pooled-null mean (6.31). [Table created by the student author using LaTeX, 2026]

The pooled synthetic null against which η is defined is a separate, larger population: 18,582 valid windows, each an independent synthetic object drawn from the fleet's regime (700–1,000 km altitude, 50–105° inclination, $e = 10^{-5}$–10^{-3}) and carried through the same TLE-convention fits (2-day cadence, ±2-day spans, B^* estimated, 5 km position-RMS gate) as the real fleet, with the same quality cuts and symmetric exclusion. Noise is added per channel as a matched deficit (the stream's one-step scatter estimated from its own residuals, floored and capped against the production covariance), so the actual and assumed covariances agree by construction. Its mean peak (6.31) far exceeds the Gate-1 intrinsic null (≈0.8, a single geometry near 800 km) because the pooled population spans the fleet's full altitude range, including the top of the band where the B^*/a fit degeneracy dominates the extreme tail: all twenty of its largest peaks sit at 891–903 km (Section 7.2). The design threshold is therefore calibrated on a population whose extreme tail is driven by a documented fitting artifact; the consequence is conservative: the artifact inflates the null tail, so η sits higher than a fleet-representative null would place it, and the fleet's real false-alarm rate at η is below the 0.1% design value (Section 6.2). Section 6.2 also reports the fleet count at every point of the percentile ladder so the sensitivity to threshold choice is visible.

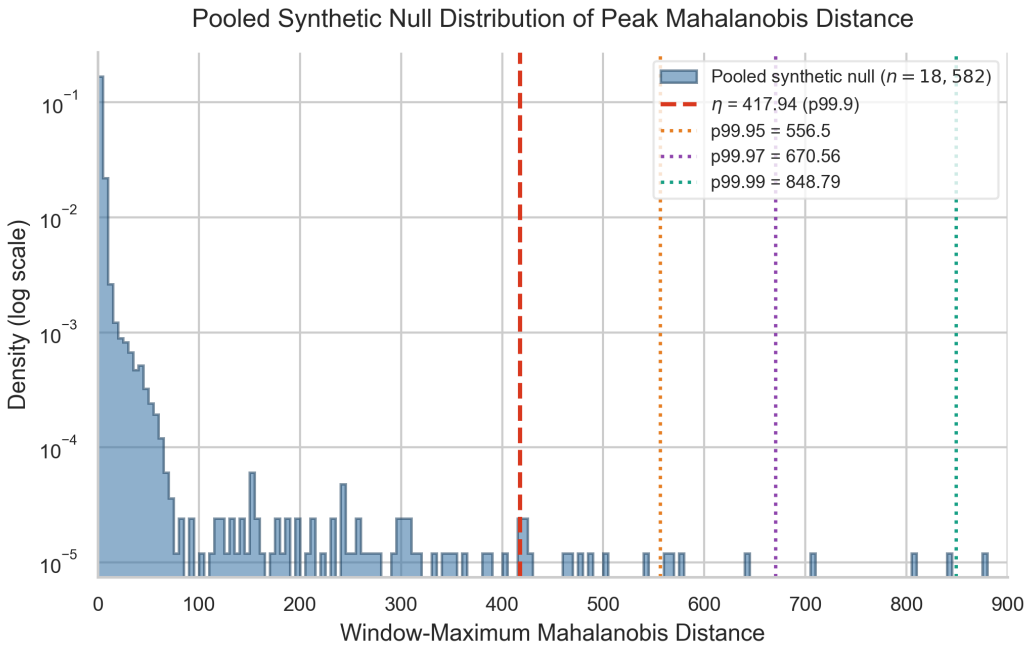


Figure 2: Null distribution of peak Mahalanobis distances (histogram with step-filled density), pooled across 18,582 valid synthetic trials. Design threshold at $\eta = 417.94$, the 99.9th percentile (0.1% per-window false-alarm rate); the ladder of upper percentiles is $p_{99.95} = 556.5$, $p_{99.97} = 670.56$, $p_{99.99} = 848.79$, max $= 1110.7$. [Graphic created by the student researcher using Matplotlib (Python), 2026]

4.3 Residual-Permutation Reference

Synthetic validation alone cannot establish reliability on real catalog data: TLE residuals contain temporally coherent structures (station changes, unmodeled maneuvers, density mismodeling) absent from the synthetic null. The residual-permutation null addresses this gap. For each object, the one-step secular-prediction residuals (the same quantities from which the per-object covariance of Section 3.2 is estimated) are randomly permuted in time, and the stream rebuilt as a chain: each step advances the previous scrambled state with the Brouwer secular rates and adds a permuted original residual, with angles wrapped and eccentricity clamped (Ernst, 2004). Permutation preserves the residual amplitude distribution, per-channel noise statistics, and cross-channel correlation, while destroying temporal coherence and any genuine dynamical signatures; by construction the injected noise scale equals the empirical per-channel scatter, so the null and the filter agree on what the stream's noise is. The same symmetric exclusion is applied; the permuted population diverges at 2.26% (a consequence of its re-fit construction at the top of the modeled band; Section 7.2) while the fleet diverges at 0.44%, and all denominators are post-exclusion.

The permuted null serves two roles. First, it is a scale check on the synthetic calibration at a matched percentile: at the 99.97th percentile, the permuted null sits at 370.7 and the observed fleet's own tail at 426.6, agreeing within ≈15%; the pooled synthetic null sits at 670.56, a factor of ≈1.6–1.8 above both, inflated by the top-of-band B^*/a artifact, making η conservative for the fleet. Second, anything temporally coherent in the original histories (genuine impulses, sustained mismodeling, residual maneuvers) is excluded from the null by design, and therefore appears as excess in the fleet comparison. The residual-permutation reference thus provides an independent and arguably more realistic non-debris background; Section 6.2 reports the

fleet count against both it and the pre-specified synthetic design rate.

4.4 Estimator Characterization and Decision Criterion

The bias-recovery tests confirm that a constant acceleration of 10^{-6} m/s^2 is tracked without generating false triggers. The FOGM parameters $\tau = 3 \times 10^4$ and $q = 10^{-19}$ were tuned via a binary grid-search that converged at these values, then also tested for impulsive event sensitivity (Section 5) to ensure impulses were not absorbed by the FOGM. The detection threshold is $\eta = 417.94$, the 99.9th percentile of the pooled synthetic null distribution (18,582 valid trials), a design false-alarm rate of 0.1% per window; with 15,760 valid fleet windows, this calibration expects $15{,}760 \times 0.001 = 15.76$ noise-only triggers, the background against which the fleet count of Section 6 is compared. The pre-specified model-discrimination criterion requires $|\log_{10} B| > 1$ to declare decisive statistical evidence between competing hypotheses (Kass & Raftery, 1995): detection efficiency $\epsilon(\Delta v)$ and measured false-positive rates enter the expected event counts under ORDEM and MASTER, and discrimination is declared only if the difference between the models' predicted contributions is not absorbed by the unidentified systematic component, which enters as a nuisance parameter. This criterion is evaluated in Section 6.3.

5 Impulse Detectability and Recovery

5.1 Synthetic Impulse Recovery

Controlled Δv impulses were injected along the anti-velocity ($-\mathbf{S}$) axis into synthetic trajectories in the TLE-like observation environment, across a logarithmic grid of Δv values. At the strike epoch, the true anomaly is computed from the mean anomaly, and the Gauss variational equations map the impulse into instantaneous element offsets $(\Delta a, \Delta e, \Delta i, \Delta\Omega, \Delta\omega)$ applied to the truth trajectory, which is subsequently propagated under the full force model (Vallado, 2022). Recovery is declared if the peak Mahalanobis distance exceeds η within ± 1 day of the strike epoch. Because the $-\mathbf{S}$ direction maximizes the semi-major-axis response to which the detection statistic is most sensitive, the resulting $\epsilon(\Delta v)$ represents a best-case efficiency for isotropic impacts, and Δv_{50} is correspondingly an optimistic threshold.

The environment itself is first validated without injected impulse. The fitted TLE-like elements recover the truth semi-major axis to within meters, at a near-constant offset of ≈2.9 km from the secular truth state (the mean-element convention difference), with slope ≈ 0, absorbed at window initialization; the 2.6–2.8 km position fit RMS is the SGP4 representational floor, not a fit failure. Under the per-object measurement covariance, the no-impulse trigger rate is 0.0% (0 of 7000 windows; mean peak ≈ 0.8). Under population-wide covariance, the identical environment triggers 94.7% of windows in the pure-convention control and 100% with secular dynamics included, identifying measurement-covariance misspecification as a sufficient false-positive generator; the per-object calibration removes it. The convention mismatch alone is therefore not a measurable contributor to the real-data false-positive floor under the adopted covariance. The resulting efficiency curve rises sharply across the sweep range and flattens at unity, certifying at $\eta = 417.94$:

$$\Delta v_{50} = 1.19\ \mathrm{m\,s^{-1}}, \qquad \Delta v_{90} \approx 1.7\ \mathrm{m\,s^{-1}}. \tag{7}$$

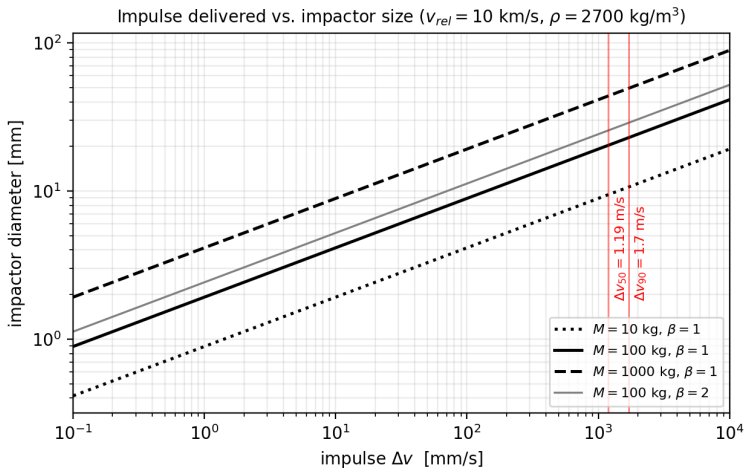


Figure 3: Impulse delivered by a hypervelocity impact as a function of impactor diameter, for spacecraft masses of 10, 100, and 1000 kg at a 10 km/s relative velocity (perfectly inelastic, $\beta = 1$, solid black curves; $\beta = 2$ momentum enhancement shown in gray for the 100-kg case) (Christiansen, 2009). Vertical lines mark the certified survey sensitivity $\Delta v_{50} = 1.19$ m/s and $\Delta v_{90} \approx 1.7$ m/s. [Graphic created by the student researcher using Matplotlib (Python), 2026]

5.2 Real-TLE Impulse Recovery

The operative sensitivity certification is performed on real TLE histories, so it carries the observational noise and catalog artifacts the survey actually faces. Here the strike is injected analytically at the measurement level: the element vector is interpolated to the strike epoch, and the Gauss variational equations map the in-track impulse into element offsets added to every TLE-derived measurement from the strike epoch onward. One valid window is selected at random per object, with the strike placed at the window midpoint; recovery is declared if the window's peak Mahalanobis distance exceeds η, with no epoch localization applied, so at negligible Δv the measured rate is the window's background trigger rate. Because the measurements themselves are perturbed and no dynamics are run after the strike, this variant preserves the real cadence, noise, and data gaps of the catalog history while embedding a known impulse.

The sweep at $\eta = 417.94$ (150 paired trials per Δv point) reports $\Delta v_{50} = 1.21\ \mathrm{m\,s^{-1}}$, with the efficiency ladder rising from $\epsilon = 0$ below ≈0.8 m/s through $\epsilon = 0.886$ at 1.6 m/s to $\epsilon = 0.965$ at 3.2 m/s (Figure 4). The response is well described by a mean peak statistic of $\approx 273\,\Delta v$ (m/s) plus a baseline floor: the crossing is signal-dominated, with the mean peak rising $218 \rightarrow 435 \rightarrow 877$ across 0.8–3.2 m/s, so Δv_{50} is insensitive to the floor. Because the in-track, mid-window strike is the most detectable geometry, this $\epsilon(\Delta v)$ is a best-case efficiency and Δv_{50} an optimistic threshold; real impacts at arbitrary geometry are less detectable per unit Δv, so the predicted detectable counts of Section 5.4 are upper-bound-ish and the size horizon below is a lower bound.

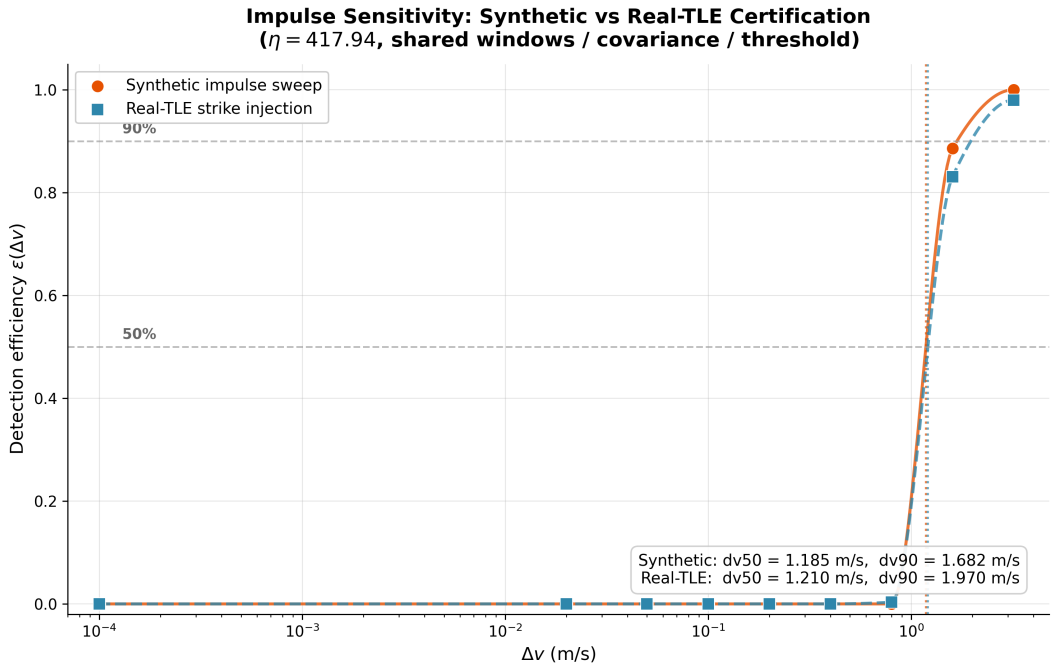


Figure 4: Detection efficiency $\epsilon(\Delta v)$ for the impulse sweep at the design threshold $\eta = 417.94$: $\epsilon = 0$ below ≈0.8 m/s, $\epsilon = 0.886$ at 1.6 m/s, $\epsilon = 1.0$ at 3.2 m/s, with $\Delta v_{50} = 1.19$ m/s; the measured negligible-Δv baseline is 0.0% (0 of 149 anchor trials). The real-TLE efficiency $\epsilon_{\text{real}}(\Delta v)$ at the same impulses and threshold is shown alongside: $\epsilon = 0.886$ at 1.6 m/s, $\epsilon = 0.965$ at 3.2 m/s, with $\Delta v_{50} = 1.21$ m/s. [Graphic created by the student researcher using Matplotlib (Python), 2026]

At negligible Δv the real-TLE curve reduces to the window background trigger rate: 0 of 149 anchor trials exceed η (one-sided 95% upper bound ≈2.0%). This is consistent with both the design false-alarm rate (≈0.1% per window) and the full-fleet trigger rate (0.038%), each of which predicts fewer than one trigger in 149 trials (≈0.15 and ≈0.06). The anchor test therefore confirms that the floor is small but cannot resolve it below ∼2%. Because it uses the same windows, covariance, and threshold as the fleet run, it is a consistency check, not an independent estimate. The injection test also samples one window per object whereas the fleet statistic weights windows by count, so the two rates are not strictly comparable.

5.3 Detection Horizon in Fragment Size

The certified sensitivity translates directly into a minimum detectable fragment size. At head-on closure (≈10 km/s), $\Delta v_{50} = 1.19$ m/s on a 200-kg spacecraft corresponds to a fragment momentum of ≈ 240 kg m/s, i.e. a ≈ 24 g body, or a characteristic length

$$L_{c,\min} \approx 2\text{–}3 \text{ cm} \tag{8}$$

(Figure 3) (Johnson et al., 2001). The survey is therefore blind to the 1 mm–1 cm population, which imparts only $\sim 10^{-5}$–10^{-1} m/s on a 200-kg spacecraft, two to three orders of magnitude below Δv_{50}, and first becomes sensitive in the uncatalogued 1–10 cm band. In physical terms, $\Delta v_{50} = 1.19$ m/s in-track corresponds to a semi-major-axis step of $\Delta a = 2a\,\Delta v/v \approx 2.3$ km at 800 km altitude ($v \approx 7.45$ km/s): the detector sees only km-scale changes in a, comparable to the ≈2.9 km TLE-fit convention offset (Section 5.1) and the fit excursions that dominate the null's extreme tail (Section 7.2). The sensitivity floor

and the calibration's dominant artifact therefore live on the same physical scale, which both explains the weak sensitivity and is the reason the threshold must be set empirically rather than analytically. This is the regime where the model comparison of Section 6.3 carries physical leverage.

5.4 Expected Observable Signals Under ORDEM and MASTER

The raw debris flux predictions from ORDEM and MASTER cannot directly be compared with the observed trigger population; each model's fragment distribution must be propagated by IMPULSE's selection function. The flux predictions are interpolated into each object's orbital regime and propagated through IMPULSE's fragment-to-impulse mapping and measured detection efficiency (Braun, Horstmann, et al., 2021; Matney et al., 2019). The predicted counts are conditional scenario predictions valid under the following input assumptions: $1.0\,\mathrm{m}^2$ cannon-ball cross-section; nominal $M_{\mathrm{sat}} = 200\,\mathrm{kg}$; fragment sizes and masses from the NASA Standard Breakup Model (Johnson et al., 2001; Reynolds et al., 1998); perfectly inelastic momentum transfer ($\beta = 1$); impact velocities consistent with each object's inclination (Christiansen, 2009); and the certified sweep efficiency curve at $\eta = 417.94$. For object j,

$$N_{\mathrm{det},j} = F_j A T_j \bar{\epsilon}_j, \qquad \bar{\epsilon}_j = \mathbb{E}[\epsilon(\Delta v)]\,, \qquad \Delta v = \frac{m(L_c)}{M_{\mathrm{sat}}}\, v_{\mathrm{rel}}, \tag{9}$$

where F_j is the interpolated cumulative flux (impacts m^{-2} yr^{-1}), A the effective cross-section, T_j the object's searched exposure, and the expectation is estimated by Monte Carlo sampling (10^5 draws per object, fixed seed), restricted to the size band in question. Summing over the fleet yields the predictions of Table 3. Because Eq. (9) is linear in F and A, the predicted counts inherit those uncertainties directly; the only strongly nonlinear input is M_{sat}, held at its nominal value. The predictions are therefore reported as conditional scenarios rather than precision forecasts.

Band	Model	Expected impacts	Expected detected	Effective ε
1 mm–1 cm	ORDEM	5,093	0.000	0.0000
	MASTER	36.7	0.000	0.0000
1–10 cm	ORDEM	0.404	0.0052	0.0129
	MASTER	0.832	0.0106	0.0128

Table 3: Predicted detections by model and size band. Detected counts use the searched exposure of 3,061.8 object-years (3,013 flux-matched objects; full-fleet reference 3,279.3 object-years; the remaining 312 objects contribute windows but no flux prediction). The vanishing 1 mm–1 cm prediction is a property of the survey's certified sensitivity (no fragment in that band reaches $\Delta v \approx 1.19$ m/s), not a claim about the flux itself: the band's expected-impact totals (5,093 vs. 36.7) preserve the models' $\sim$138$\times$ flux divergence, but none of those impacts is detectable. [Table created by the student author using LaTeX, 2026]

The exposure T_j is the same searched exposure over which observed triggers are counted, so the predicted and observed counts share a denominator by construction. The efficiency applied is the certified sweep curve of Section 5.2, whose negligible-Δv floor is the window background rate; because the sweep efficiency is a best-case recovery, the predicted counts are upper bounds on detectable events under each model.

In the 1 mm–1 cm band, the two models disagree by $\sim$138$\times$ in flux (ORDEM grid fluxes of 0.48–2.53

vs. MASTER's 0.003–0.018 m^{-2} yr^{-1}). The survey is blind to this band by impulse scale, so the bracket is reported without privileging either model, and the band contributes no predicted detectable events under either model; it carries no decision weight. In the 1–10 cm band (the band that matters for this survey, since it is where the detection horizon opens), the two models *agree* within a factor of ≈3 (ORDEM 4.8×10^{-5}–1.6×10^{-4}; MASTER 9.2×10^{-5}–5.1×10^{-4} m^{-2} yr^{-1}; ORDEM/MASTER ratio 0.31–0.79, MASTER higher), making the model-dependent conclusion robust. Table 3 reflects this structure: MASTER predicts ≈2× more expected impacts than ORDEM at nearly identical effective efficiencies (0.0129 vs. 0.0128), so the ratio of predicted detections (2.0; ORDEM/MASTER 0.49) falls inside the models' own flux-ratio bracket, a consistency check that the selection-function machinery preserves the relative fluxes it is fed. The efficiencies are largely similar because the dominant contributor, the debris fragmentation model, is the same NSBM across the two. The significantly low efficiencies and predicted detections also highlight IMPULSE's limits: it is not suitable for standalone event debris detection, but rather a discrimination pipeline.

6 Fleet-Scale Analysis

6.1 Fleet Characteristics and Raw Detection Statistics

The complete pipeline was executed on this set via memory-safe multiprocessing; summary statistics are in Table 4. The altitude distribution peaks around 820 km and 880 km; 55% of the objects occupy the sun-synchronous inclination shell ($95°$–$105°$), and observation spans range from a single valid window to 17 years of continuous coverage (Figure 5), consistent with the SSO regime's widespread commercial use.

Metric	**Value**
Total objects processed	3,325
Valid analysis windows (pre-exclusion)	15,830
Divergent windows excluded (symmetric policy)	70 (0.44%)
Valid analysis windows (post-exclusion)	15,760
Mean peak Mahalanobis distance	8.68
Design threshold η (99.9^{th} percentile, synthetic null)	417.94
Noise-only expected triggers at η	15.76
Raw triggers ($> \eta$)	6
Raw trigger rate	0.038%

Table 4: Full fleet run summary statistics. [Table created by the student author using LaTeX, 2026]

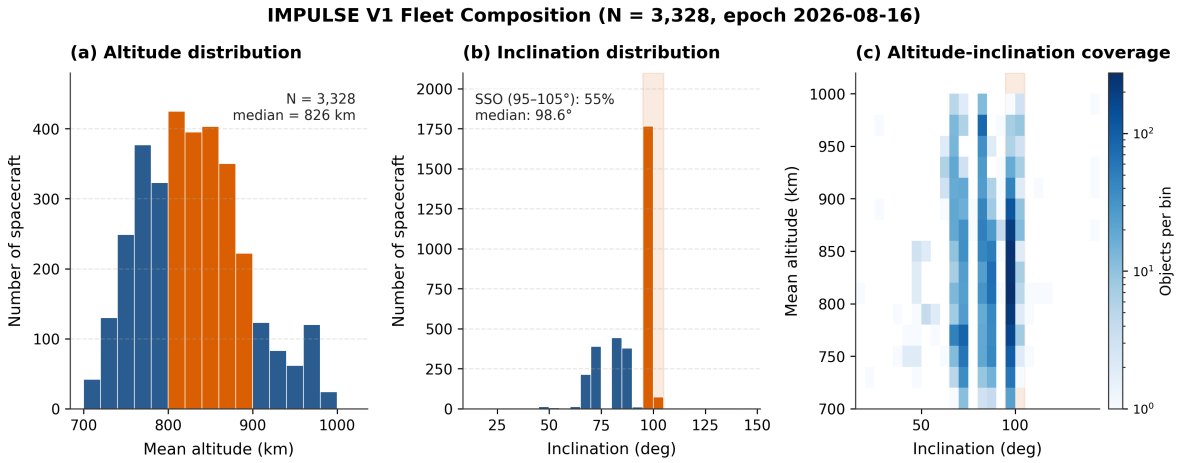


Figure 5: IMPULSE fleet composition at the final catalog epoch (2026-08-16). (a) Mean-altitude distribution; all objects lie within the 700–1000 km selection band. (b) Inclination distribution; the sun-synchronous shell (95–105°, orange) contains 55% of the fleet. (c) Joint altitude-inclination coverage (log color scale). [Graphic created by the student researcher using Matplotlib (Python), 2026]

When the calibrated pipeline is applied to the 15,760 valid windows, 6 peaks exceed $\eta = 417.94$, a raw trigger rate of 0.038%. The mean peak Mahalanobis distance across all valid windows is 8.68, against 6.31 for the synthetic null (≈38% higher), and the distribution is strongly right-skewed, with a fleet 99.97th percentile of ≈426.6, so a small number of extreme windows dominate the upper tail. The mean offset is a mismatch in distribution shape between fleet and null, not a validation of the calibration: below roughly the 99th percentile the fleet trigger rate runs above the null-only expectation, and only in the extreme tail (from the 99.5th percentile upward, including the design point at 99.9) does the fleet fall to or below it. All conclusions below are drawn at the design threshold, above this crossover, where the mismatch works against rather than for the survey, but the same mismatch means the fleet's true background at η is below the 0.1% design rate, which limits the power of the excess test (Section 6.2). As a calibration ablation, the same fleet was processed under the population-wide published covariance: at a fixed threshold the trigger rate collapses from 51.55% to 18.75%, a 2.75× reduction identifying measurement-covariance misspecification as the dominant fleet-scale false-positive generator. The per-object calibration removes the mechanism entirely in the synthetic environment (94.7% → 0%, Section 5.1), but on the fleet it only reduces it: the residual 18.75% at that fixed threshold is absorbed by the empirically calibrated η rather than eliminated, so the fleet claim is one of reduction, not removal.

NORAD ID	Peak η	Equivalent Δv (m/s)	Ref. L_c (cm)
31949	3972	≈14.5	≈6 (lower bound)
39778	936	≈3.3	≈3.5
30579	574.6	≈2.0	≈3.0
35065	574.2	≈2.0	≈3.0
35284	436.1	≈1.5	≈2.7
30408	423.1	≈1.5	≈2.7

Table 5: Trigger candidates at the design threshold. Equivalent impulses assume the in-track, mid-window, single-strike inversion of the certified sweep (Section 5.2); NORAD 31949's peak lies beyond the calibration grid and is quoted as a lower bound. Reference impactor sizes L_c scale the certified $\Delta v_{50} \rightarrow L_c$ mapping of Section 5.3 (head-on, 10 km/s, $M_{\mathrm{sat}} = 200$ kg, $\beta = 1$) by the equivalent Δv; they are indicative scales, not attributions. All six candidates remain unattributed: no public conjunction, maneuver, or breakup association was established. [Table created by the student author using LaTeX, 2026]

The six triggering windows are isolated events on six distinct objects, spread across the fleet rather than concentrated in any same-launch cluster or single epoch. Their equivalent impulses, 1.5–15 m/s under the single-strike inversion, sit two to three orders of magnitude above anything the 1 mm–1 cm population can produce, and within the range that only 3–6 cm head-on fragments reach (Table 5). Against the pooled null, five of the six candidates (peak η from 423.1 to 936) lie within the range of the null's extreme tail (null maximum 1110.7), but NORAD 31949 (peak $\eta = 3972$, a factor of 3.6 above the null maximum) is not consistent with the calibrated tail: it is the one window for which a noise origin is unsupported by the present calibration, though it remains unattributed.

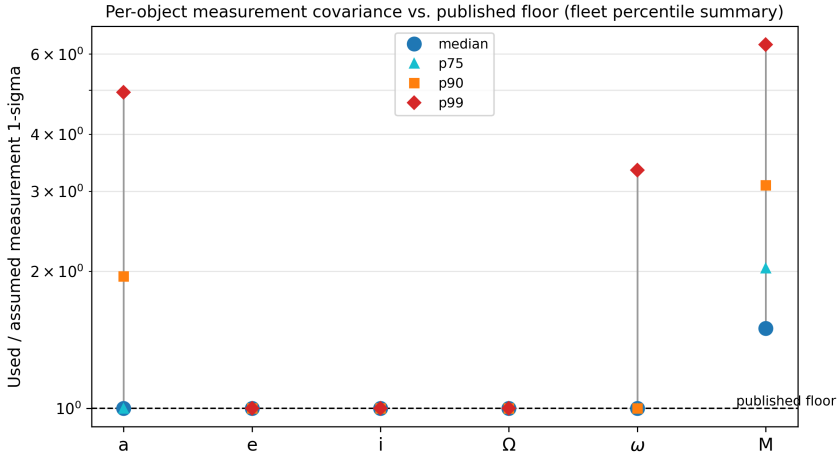


Figure 6: Per-channel ratio of the used (per-object, empirically estimated) measurement 1-sigma to the assumed (published-floor) 1-sigma across the fleet: median (bold marker) with the 75th, 90th, and 99th percentiles per channel. The median object sits near the published floor ($1.0\times$) on most channels and at $\approx 1.5\times$ on M; the widening acts on a minority of noisier objects, which is where the false positives originate (Sections 3.2 and 6.1). [Graphic created by the student researcher using Matplotlib (Python), 2026]

6.2 Background Subtraction and the Net Excess

The background subtraction can be performed by direct difference. The noise calibration expects 15.76 triggers among the 15,760 valid windows; 6 are observed:

$$N_{\text{net}} = N_{\text{obs}} - N_{\text{noise}} = 6 - 15.76 = -9.76 \pm 4.7 \tag{10}$$

with $\sigma^2_{\text{net}} = N_{\text{obs}} + (N_{\text{valid}}\sigma_r)^2$ and $\sigma_r = \sqrt{r(1-r)/N_{\text{valid}}} = 0.0252\%$ for the calibrated rate $r = 0.001$, combining Poisson counting fluctuation with the finite-sample binomial uncertainty of the rate. The subtraction is performed on aggregate counts, not by pairing individual windows, and the uncertainty is statistical only; because the excess is reported descriptively, unmodeled dependence terms do not affect the conclusions. Table 6 compares the observed count against the noise-plus-model predictions: all five scenarios predict totals of 15.76–15.77 against the 6 observed, so the model-predicted debris contributions, 0.000033% (ORDEM) and 0.000067% (MASTER) of windows in the 1–10 cm band and nothing in the 1 mm–1 cm band, are negligible against both the background and the observed count.

Scenario	**Predicted total**	**Observed**	**Residual**
Noise only (0.1%/window)	15.76	6	-9.76
Noise + ORDEM (1–10 cm)	15.765	6	-9.765
Noise + MASTER (1–10 cm)	15.771	6	-9.771
Noise + ORDEM (1 mm–1 cm)	15.76	6	-9.76
Noise + MASTER (1 mm–1 cm)	15.76	6	-9.76

Table 6: Observed triggers against noise-plus-model predictions. Rows use E_{det} from the 1 mm–1 cm band (no detectable events under either model, by sensitivity) and the 1–10 cm band (0.005 ORDEM, 0.011 MASTER); predicted totals are $15.76 + E_{\text{det}}$. [Table created by the student author using LaTeX, 2026]

The observed count lies *below* the noise-only expectation: the probability of observing six or fewer noise-only triggers is $P \approx 0.005$ (one-sided Poisson). This deficit is defined against the synthetic design rate, not the permuted null of Section 4.3: the permuted population's 99.97th percentile (370.7) lies below η, bounding its false-alarm rate at η below 0.03% per window (fewer than ≈4.7 expected triggers), against which the observed six would be a surplus rather than a deficit. The exact permuted rate at η was not tabulated, so the sign of the fleet–null residual is reported as null-dependent: the design-rate comparison is retained as the pre-specified background, and the robust conclusion carried forward is that no positive excess is certifiable against either null. Taken at face value, a deficit means the null does not fully describe the fleet: the fleet's true false-alarm rate at η is below the 0.1% design value, so η is conservative and the survey's power to detect a small excess above the true background is limited. The "no certifiable excess" verdict should be read jointly with that low power: it certifies the absence of a large excess, not the absence of any signal. The excess upper limit is a one-sided 95% Poisson limit on the signal mean μ above background b. Under the calibrated background $b = 15.76$, $P(N \leq 6 \mid 15.76) = 0.0047 < 0.05$ already at $\mu = 0$, so $\mu_{\text{UL}} = 0$: conditional on the calibrated background, the 95% upper limit on excess triggers is zero. The limit's background dependence is stated explicitly: it remains zero even if the true background is half the design rate ($b = 7.88$ gives $P = 0.33$), and only if the true background were as low as the observed

count itself ($b \approx 6$) would the limit open, to $\mu_{\rm UL} \approx 5.8$; under the permuted null's lower background ($b < 4.7$) the same construction opens to $\gtrsim 7$ excess events. A fuller treatment would use a unified frequentist construction with the background as a nuisance parameter (Feldman & Cousins, 1998). The present limit is reported as conditional on the calibration.

There is therefore no positive excess to decompose at the design threshold. This is the central fleet-scale result: the survey certifies no excess of temporally coherent triggers above the calibrated noise background, five of the six candidates lie within the calibrated tail, and one (NORAD 31949) is not covered by it. The conclusion is robust across the percentile ladder of Figure 2: at $p_{99.95}$ the fleet count is 2 against 7.9 expected, at $p_{99.97}$ it is 1 against 4.7, and at $p_{99.99}$ it is 0 against 1.6, at or below the noise-only expectation at every point of the ladder. Potential contributors to the residual deficit and the individual extreme windows include catalog discontinuities, residual atmospheric and environmental mismodeling, and maneuvers by objects incorrectly classified as non-maneuvering; the convention discrepancy is quantified in Section 5.1 and contributes a ≈0% floor under the adopted covariance. Their individual contributions are not separately quantified.

6.3 Comparison With ORDEM and MASTER, and Verdict

When the fleet measurement is compared with the selection-function-propagated predictions from Section 5.4, the outcome is unambiguous. The observed count (6) lies below the noise-only expectation (15.76), so there is no excess to attribute to debris under either model. The predicted detectable-impact counts, none in the 1 mm–1 cm band and 0.005 / 0.011 (ORDEM / MASTER, 1–10 cm), are far below the six candidates and far below the noise background itself; the candidates are therefore classified as unattributed threshold exceedances, consistent with the noise model, and the survey certifies zero debris-impact detections. This is the regime in which the background explains the triggers, and discrimination between the models requires flux-ratio leverage and improved sensitivity: the models' fluxes differ by ∼138× in the 1 mm–1 cm band (where the survey is blind by impulse scale), but their predicted detectable counts differ by only a factor of ≈2 in the 1–10 cm band, so even a positive excess could not have discriminated between them at the present sensitivity.

The pre-specified $|\log_{10} B| > 1$ criterion of Section 4.4 is therefore not met: with zero attributable detections, the evidence factor between the ORDEM and MASTER hypotheses is not decisive, and no discrimination verdict is declared. Because none of the six candidates is attributable to debris, the detectable-impact count is zero, and the standard zero-count one-sided Poisson limit ($e^{-\mu} = 0.05$) gives a 95% upper limit of ≈3.0 detectable events over the fleet exposure; the $\geq L_{c,\min}$ (≈2–3 cm) impact rate is correspondingly bounded at no more than $\approx 3/E_{\rm det}$ times the model-predicted rate. This limit is conditional on classifying all six candidates as noise: were they instead counted as debris, the observed-count Poisson limit would be ≈11.8 events and the rate bound would weaken by a factor of ≈4 (to ≈2,300× ORDEM and ≈1,100× MASTER). The classification is motivated — five of six candidates lie within the calibrated tail (Section 6.1) — but the dependence is stated rather than hidden. The strength of this bound must be stated plainly: with $E_{\rm det} = 0.005$ (ORDEM) and 0.011 (MASTER) in the 1–10 cm band over the same exposure as the observed count, the ≈3-event limit caps the ≥ 2–3 cm impact rate at $\approx 570\times$ (ORDEM) and $\approx 280\times$ (MASTER) the model-predicted rate, a factor of hundreds, not a discrimination. The bound is honest but weak, consistent

with the paper's own logic that discrimination was never possible at this sensitivity. This is not evidence that either model is incorrect: the observed triggers must be interpreted strictly as an upper bound on the combined debris-plus-systematic rate rather than as a measurement of the debris population itself.

7 Discussion

7.1 What IMPULSE Demonstrated

IMPULSE extends the dual-EKF + Fraser-Potter + FOGM architecture explored by Bennett et al. (2023) to fleet-scale historical TLE mean-element analysis. Controlled tests demonstrate sensitivity to injected impulses in a TLE-like environment whose no-impulse floor is 0.0%, and the sweeps certify $\Delta v_{50} = 1.19$ m/s in the synthetic environment and 1.21 m/s on real TLEs: a 2–3 cm head-on fragment on a 200-kg spacecraft, a population for which, to the author's knowledge, the survey provides the first TLE-based passive-sensing constraint. The principal finding on the false-positive side is that the dominant fleet-scale false-positive mechanism is measurement-covariance misspecification, which per-object empirical calibration reduces (51.55% to 18.75% at a fixed threshold). The TLE mean-element convention mismatch, quantified end-to-end for the first time in this pipeline, is benign under the adopted covariance. The calibrated fleet measurement (six triggers against 15.76 expected, $P \approx 0.005$) leaves no certifiable excess: the survey certifies zero debris-impact detections and sets a 95% upper limit of ≈3 detectable events over the fleet exposure, fully consistent with both models' predictions.

7.2 Limitations and Path Forward

After per-object calibration, the fleet tail behaves consistently with the null, and the tested synthetic confounders do not reproduce the trigger rates seen under miscalibrated covariance; however, the relative contributions of catalog effects, observation-model mismatch, and other unmodeled processes to the six extreme windows remain unresolved, and the candidates are left unattributed. The specific limitations of the present implementation are:

1. **Calibration artifact.** The B^*/semi-major-axis degeneracy at the band top drives the null's extreme tail (all twenty largest null peaks sit at 891–903 km), so η is calibrated partly against a documented construction artifact. The consequence is conservative, but η should not be reused for another fleet without recalibration.
2. **Statistical approximations.** Same-object windows share covariance and residuals, making the Poisson variance approximate; the zero-excess limit is conditional on the calibrated background: the fleet–null shape mismatch both opens the limit ($\mu_{\mathrm{UL}} \approx 5.8$ if the background were as low as the observed count) and lowers the test's power.
3. **Best-case geometry and floor.** Δv_{50} was calibrated on the most detectable strike geometry with an elevated anchor floor (mean peak ≈22.3 vs. null 6.3 at negligible Δv); the crossing is signal-dominated, so expected-detectable counts are upper-bound-ish and $L_{c,\min}$ a lower bound.
4. **Coverage and inputs.** 2026-epoch fluxes are applied across the 2009–2026 window; ≈58% of flux-matched objects receive extrapolated fluxes; the diagonal per-channel R cannot represent a/M correlation; the per-object covariance is estimated from the tested stream itself; the synthetic validation

covers a single geometry near 800 km; and 312 fleet objects carry no flux prediction, a one-sided gap that can only understate the predicted counts.

Therefore, the path forward is improved sensitivity and exposure rather than statistical power. The detectable population ($\gtrsim$ 2–3 cm) is too rare for discrimination: both models predict 0.005–0.011 events over the present exposure, requiring longer exposure, a larger fleet, or lower $\Delta v_5 0$. (through improved force modeling, full-covariance measurement updates, and cross-channel detection statistics) to bring the 1–10 cm population within reach. Additional diagnostics should stratify triggers by object type (Li & Chen, 2018) and test associations with geomagnetic activity and the solar cycle, constraining the unidentified component before attempting debris-model discrimination.

8 Conclusion

IMPULSE tested whether historical TLEs of inactive LEO spacecraft can support debris-environment model discrimination between NASA's ORDEM 3.1 and ESA's MASTER 8, which disagree by $138\times$ in the orbital debris flux between 700–1000 km altitudes. The measurement covariance is calibrated per object from each stream's one-step prediction residuals, and the TLE mean-element convention mismatch is quantified directly by an end-to-end synthetic experiment: it contributes a 0.0% false-positive floor under the adopted covariance. The design threshold $\eta = 417.94$ is certified against a pooled synthetic null of $18,582$ trials; at a matched 99.97^{th} percentile, the residual-permutation null (370.7) and the fleet's own tail (426.6) agree within $\approx 15\%$, while the synthetic null sits a factor of ≈ 1.6–1.8 higher, making the threshold conservative for the fleet. The certified sensitivity is $\Delta v_{50} = 1.19$ m/s (1.21 m/s on real TLEs): a $\approx$ 2–3 cm head-on fragment on a 200-kg spacecraft. Applied to a fleet of $3,325$ decommissioned spacecraft ($15,760$ valid analysis windows), the pipeline produced 6 trigger candidates (0.038%) against 15.76 expected from the noise calibration alone: no positive excess is certifiable ($P \approx 0.005$ against the design rate, and a count that is instead a mild surplus against the permuted null's lower background; 95% upper limit on excess ≈ 0 under the design background), the six candidates remain unattributed, and the models predict no detectable events in the 1 mm–1 cm band and $0.005/0.011$ in the 1–10 cm band over the same exposure. The survey certifies zero debris-impact detections, consistent with the models (which agree within $\approx 3\times$ in the decision-relevant 1–10 cm band), and constrains rather than claims: the $\geq$ 2–3 cm impact rate is bounded at $\approx$ 280–570$\times$ the model-predicted rate, insufficient to discriminate between models. Further work must lower Δv_{50} and extend the exposure before these anomalies can constrain the debris environment.

Data, Code, and AI-Use Statement

All TLE data came from Space-Track. The IMPULSE pipeline and analysis scripts are hosted on GitHub and available from the author upon request. Claude (Anthropic) assisted with the Figure 1 schematic, several figures and their Matplotlib scripts, and supplemental code (argparse logic and file reorganization); the author modified, vetted, and corrected these contributions. All other code, analyses, results, conclusions, and decisions are the author's own. The report was written solely by the author, with AI used only as a learning aid for LaTeX and Overleaf.

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